RBC CAPITAL MARKETS · DIGITAL ASSET TECHNOLOGY

Hyperliquid 101

How a fully on-chain order book exchange works: architecture, the money in, the trade, the risk engine and the money out, explained in plain English.

0 Hyperliquid at a glance

1 The mental model

2 Core architecture: two engines

3 How assets enter

4 How a trade happens

5 Liquidation and the HLP vault

6 How assets exit

7 Risks and trade-offs

8 Questions, verification notes, glossary



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SUMMARY FIRST

0 Hyperliquid at a glance

Hyperliquid is a decentralized perpetual futures and spot exchange that runs as its own Layer-1 blockchain. The simplest way to picture it is a centralized exchange matching engine whose internal database has been replaced by a fast, consensus-driven state machine that every validator runs.[1]

~0.2 s
Finality / median end-to-end order latency[1,2]

200,000
Orders per second, stated capacity[1,2]

Every 3 s
Validators publish oracle prices[1,3]

Hourly
Funding settled peer-to-peer (vs 8-hourly on most CEXs) [1,4]

What it is — OVERVIEW
An exchange for perpetual futures ("perps") and spot tokens, running on its own purpose-built blockchain rather than on Ethereum or another chain.

Design choice — OVERVIEW
Rejects both common DEX designs (AMM pools like Uniswap; hybrid off-chain matching like dYdX v3 or Vertex). Runs a central limit order book (CLOB) *inside* consensus.

Consensus — ARCHITECTURE
HyperBFT, a Byzantine-fault-tolerant proof-of-stake protocol. Blocks roughly every 100–200 ms; V3 ~0.2 s finality.

Two engines — ARCHITECTURE
HyperCore: hard-coded Rust trading engine (order books, clearinghouse, liquidations, oracle). **HyperEVM**: general-purpose smart contracts that can read from and trade into HyperCore.

Collateral — MONEY
USDC is the unit of account and margin for the whole trading engine.

Money in — MONEY
Native Arbitrum bridge (minimum 5 USDC), credited once more than 2/3 of stake attests. Also intent bridges and the Unit protocol for native BTC, ETH and SOL. Current docs now prefer native USDC via Circle CCTP. V8

Keys — ACCESS
A trade-only "agent key" places and cancels orders with no wallet pop-ups. Only the master wallet can withdraw.

Trading — TRADING
Price-time priority CLOB. Limit, Market, Stop-Market/Limit, Take-Profit, native TWAP and Scale orders. No gas per order; maker/taker fees only.

Risk engine — TRADING
Margin recalculated every block. Cross or isolated margin. Oracle = weighted median of external exchange prices, used to resist "wick" manipulation.

Default waterfall — RISK
1) Market order to the book. 2) HLP vault backstop below 2/3 of maintenance margin. 3) Auto-deleveraging (ADL) as the last resort.

Money out — MONEY
Master-wallet signature, flat 1 USDC fee, more than 2/3 of stake signs, dispute window, about 2–5 minutes in total. No ETH or HYPE needed.

Main trade-offs — RISK
Smaller, high-spec validator set than Ethereum. Bridge escrow concentrated on Arbitrum. V8 ADL can close profitable positions in tail events.

Superscript numbers in brackets refer to Sources. Gold "V" tags point to the verification notes in section 8, where the public documentation differs from or adds to the briefing text.

SECTION 1 · FRAMING

1 The mental model

Think of a normal exchange matching engine. Now take away the company's private database and replace it with a blockchain that a set of independent validators run in lock-step. That is Hyperliquid.[1]

A centralized exchange matching engine whose internal database has been replaced by a fast, consensus-driven state machine.

Three ways to build a crypto exchange

AMM (e.g. Uniswap)

No order book. Traders swap against a pool of tokens priced by a formula. Chosen because most blockchains are too slow to run an order book.

Hybrid (e.g. dYdX v3, Vertex)

A private off-chain server matches orders like a CEX, then settles results on a blockchain in batches. Fast, but matching is a black box until settlement.

Hyperliquid

A purpose-built blockchain that runs the order book itself, inside consensus. Every order, cancel and fill is a state change that all validators agree on.

Centralized exchange vs hybrid DEX vs Hyperliquid

Feature	Centralized exchange (Binance, Bybit)	Hybrid DEX (off-chain sequencer)	Hyperliquid (HyperCore L1)
Custody of funds	Exchange holds the keys (custodial)	Smart contract (self-custody)	**Validator bridge / L1 state (self-custody)**
Matching engine	Private cloud servers	Private off-chain server	**On-chain, run by every validator**
Order transparency	Black box	Opaque until settled on-chain	**Fully transparent in real time**
Gas fee per order	Free	Free (paid on batch settlement)	**Free (no gas to place or cancel)**
Latency	~5–20 ms	~10–100 ms	**~100–200 ms**

Source: briefing text.[1] The CEX and hybrid latency ranges are indicative and were not independently verified (note V12). Hyperliquid is slower than a CEX by roughly an order of magnitude but fast enough that the protocol says automated strategies can be ported with minimal changes.[2]

WHAT IT MEANS FOR CAPITAL MARKETS

- **Venue analysis, new answers.** The usual venue due-diligence questions still apply (who operates it, what the rulebook is, how defaults are managed), but the answers sit in protocol code and validator governance rather than in a regulated legal entity and a published rulebook.
- **Total transparency cuts both ways.** Every order and position is public. That helps surveillance and market-structure research, and it also means any desk or client flow placed there would be visible to everyone.
- **Speed tier.** At ~100–200 ms, it is not a venue for latency-sensitive strategies competing with co-located CEX flow, but it is far faster than most on-chain venues.

SECTION 2 · ARCHITECTURE

2 Core architecture: two engines, one chain

Hyperliquid splits its blockchain into two environments that share one state and one validator set: a hard-coded trading engine (HyperCore) and a general-purpose smart contract layer (HyperEVM).[1]

~0.2 s	finality, HyperBFT consensus
200,000	orders per second, stated capacity
0	gas to place or cancel an order

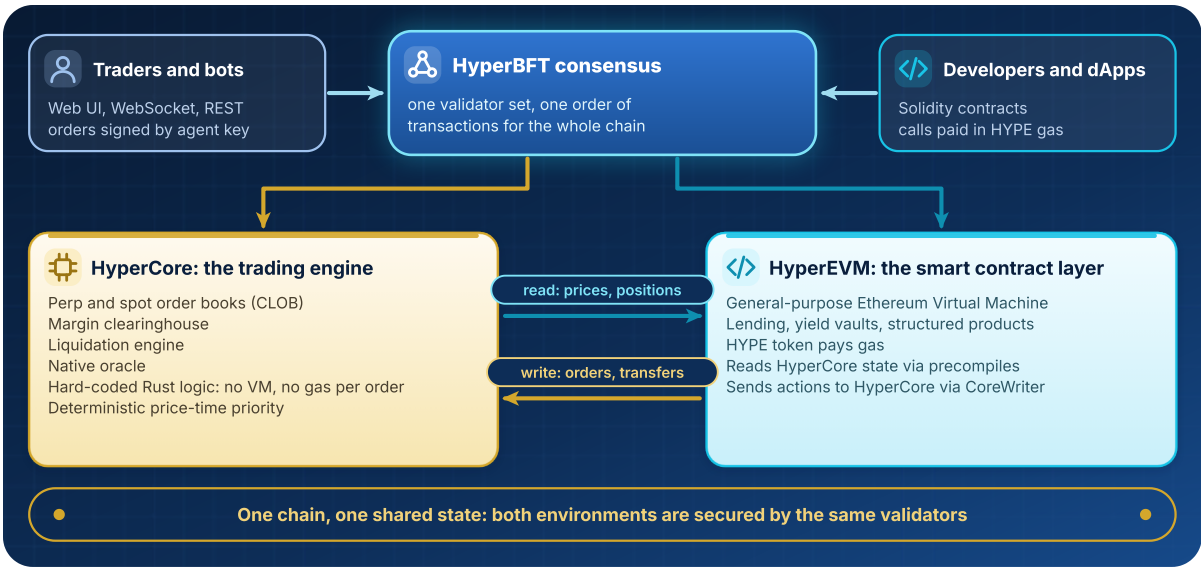


Figure 1. Orders and contract calls are sequenced by one HyperBFT consensus. HyperCore holds the exchange; HyperEVM runs programmable contracts that can read and trade into HyperCore.

HyperCore: the trading engine HYPERCORE

- **Not a smart contract.** It is Rust state-transition logic built into the chain itself.
- **What it holds:** perp and spot order books, the margin clearinghouse, the liquidation engine and the native oracle.
- **Why it matters:** skipping the virtual machine means no gas for placing or cancelling orders. Traders pay only maker/taker fees. Matching follows deterministic price-time priority.
- **Scale:** HyperBFT consensus targets ~0.2 s finality and up to 200,000 orders per second.[1,2]

HyperEVM: the smart contract layer HYPEREVM

- **A standard Ethereum Virtual Machine** attached to the same chain. Developers deploy ordinary Solidity contracts: lending, yield vaults, structured products.
- **Gas is paid in HYPE,** the chain's native token.
- **The bridge between the two:** "read precompiles" let a contract see HyperCore prices and positions. "Write" actions let it place trades into the HyperCore matching engine. The briefing says this happens in the same block; current docs say order actions sent this way are deliberately delayed by a few seconds. V9 [9]

WHAT IT MEANS FOR CAPITAL MARKETS

- **A clearinghouse in the protocol.** HyperCore's clearinghouse is the nearest analogue to a CCP risk engine, but it is run by validators and changed through protocol upgrades, not by a regulated clearing house with members and a default fund.
- **Composability is the novelty.** HyperEVM lets third parties build structured products and vaults that trade directly on the exchange's books. For product and risk teams, that means a product's risk can depend on both contract code and exchange behaviour.
- **Technology lens.** It is a useful reference design for any team thinking about putting matching logic inside a ledger rather than settling to one after the fact.

SECTION 3 · MONEY IN

3 How assets enter

Hyperliquid is its own sovereign chain, so it cannot hold Ethereum or Arbitrum assets directly. Money arrives through bridges, and USDC is the collateral for everything.[1]

5 USDC minimum deposit via the Arbitrum bridge

> 2/3 of stake must attest before credit

0 wallet pop-ups per trade with an agent key

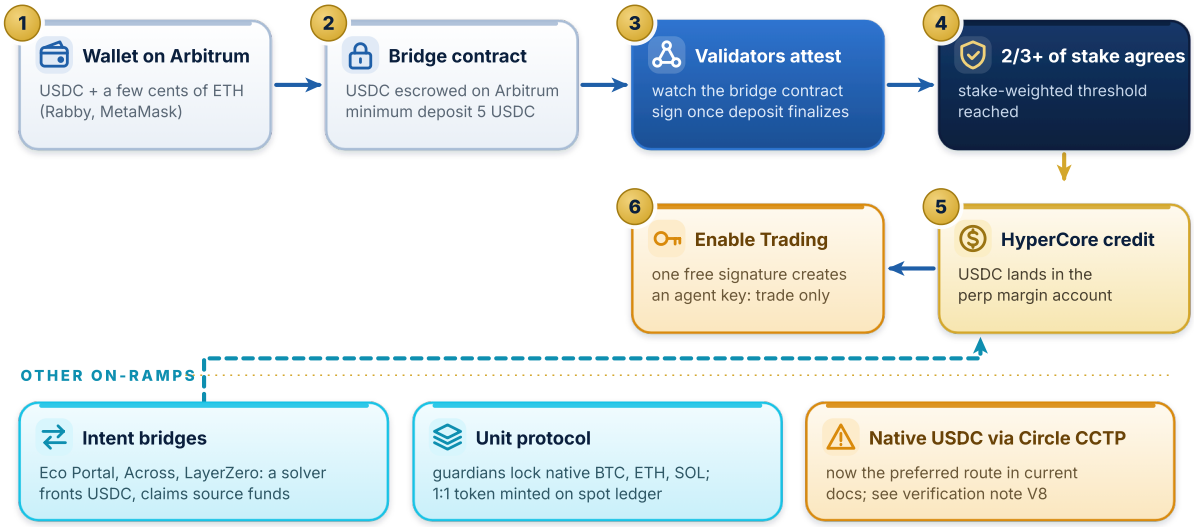


Figure 2. The primary deposit journey described in the briefing (native Arbitrum bridge), followed by the other on-ramps. The dashed line shows alternative routes arriving at the same HyperCore account.

The native Arbitrum bridge

1. **Escrow on Arbitrum.** The trader sends USDC, plus a few cents of ETH for Arbitrum gas, to the Hyperliquid bridge contract. Minimum deposit is 5 USDC.[1,8]
2. **Validator attestation.** Validators watch the contract and sign once the deposit is final on Arbitrum.
3. **Credit on HyperCore.** When more than 2/3 of stake-weighted validators have attested, the exact USDC amount is credited to the trader's HyperCore perp margin account.

Current docs call this the "legacy" bridge and describe it as deprecated in favour of native USDC via Circle's CCTP. V8 [7,8]

Other paths and the agent key

- **Intent bridges** (Eco Portal, Across, LayerZero): for USDC on Base, Solana or Ethereum. A market-making "solver" fronts the USDC instantly and claims the trader's funds on the original chain.
- **Unit protocol:** a guardian network locks native BTC, ETH or SOL on its home chain and mints a 1:1 token on Hyperliquid's spot ledger.
- **Agent wallets (session keys):** "Enable Trading" asks for one free signature that creates a local agent key (or API key for a bot). It can place, modify and cancel orders. It *cannot* withdraw. That is why trading feels like a CEX, with no wallet pop-up per trade.[1,11]

WHAT IT MEANS FOR CAPITAL MARKETS

- **Bridge and issuer exposure come first.** Before any trading risk, a participant takes on bridge risk and USDC issuer risk. Both belong in counterparty and operational risk assessments.
- **The agent-key split is familiar.** Separating trade authority from payment authority mirrors trade-only API keys and maker/checker entitlements. An institution would hold the master (withdrawal) key in institutional custody (MPC or HSM) and issue agent keys per desk or strategy.
- **Wrapped assets carry extra trust.** BTC, ETH and SOL arrive as 1:1 representations minted by a third-party guardian network. That is a different claim from holding the native asset.

SECTION 4 · EXECUTION

4 How a trade happens

Whether an order comes from the web screen or an API, it follows the same path: signed by the agent key, ordered by consensus, matched by an identical engine on every validator, and then reflected in a clearinghouse that re-checks every account every block.[1]

~100–200 ms blocks via HyperBFT

3 s oracle price updates

1 h funding settlement interval

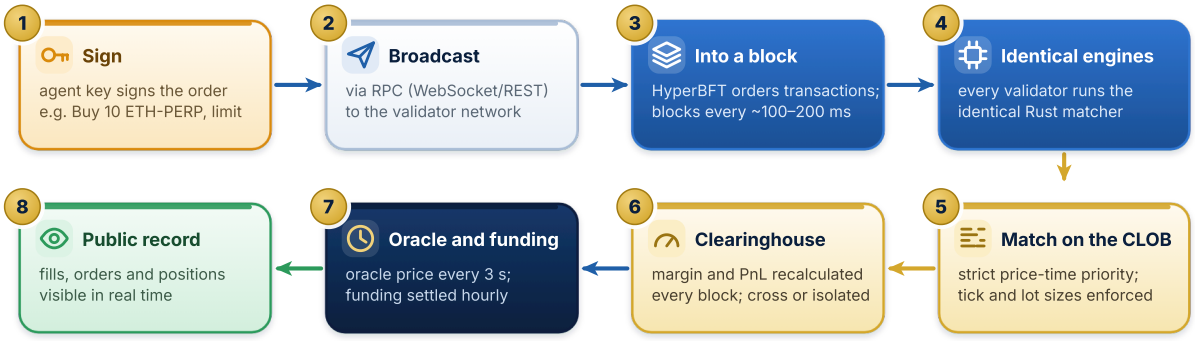


Figure 3. The life of an order. Steps 1–5 are execution; steps 6–8 are the risk and transparency layer that runs continuously.

Placement and matching HYPERCORE

- **Signing and broadcast:** the agent key signs the order (for example: Buy 10 ETH-PERP at $3,000, Limit, Post-Only) and sends it to the validators.
- **Consensus ordering:** validators bundle orders into blocks every ~100–200 ms. V3 Every validator runs the same Rust matching code in parallel, so all reach the same result.
- **Price-time execution:** an order that crosses the spread (taker) fills at once against resting orders (makers) by strict price-time priority. Tick and lot sizes are enforced by the chain.
- **Order types:** Limit, Market, Stop-Market and Stop-Limit, Take-Profit, TWAP (split natively by the chain) and Scale.

Margin, oracle and funding RISK

- **Margin modes:** all perp positions use USDC. *Cross margin* pools the perp account so gains on one position offset losses on another. *Isolated margin* ring-fences collateral to one position.
- **Oracle price:** validators feed external spot prices into the chain every 3 seconds. The weighted median drives unrealized PnL, funding and liquidation, so a manipulator cannot trigger liquidations by spiking ("wicking") Hyperliquid's own book. V7 [1,3]
- **Hourly funding:** funding is calculated and settled peer-to-peer every hour, from the gap between the perp mark price and the external oracle price. Most CEXs settle every 8 hours.[1,4]

WHAT IT MEANS FOR CAPITAL MARKETS

- **Margin is continuous, not daily.** Positions are re-margined every block and funding settles hourly. Compared with daily variation margin in listed and cleared derivatives, liquidity needs are intraday and constant, which matters for treasury and collateral operations.
- **The oracle is part of the rulebook.** Liquidation and funding outcomes depend on which external exchanges feed the oracle and how they are weighted. Model Risk and Market Risk would want that methodology documented the way an index methodology is.
- **Deterministic matching is auditable.** Because every validator re-runs the same engine, execution quality and best-execution analysis can be reconstructed from public data.

SECTION 5 · RISK ENGINE

5 Liquidation and the HLP vault

In most of DeFi, outside "keeper" bots call contracts to seize collateral, and this can jam the chain in a crash. Hyperliquid builds liquidation into the protocol and uses a community vault, HLP, as the backstop.[1]

2/3 of maintenance margin: backstop threshold

3 steps book, HLP vault, then ADL

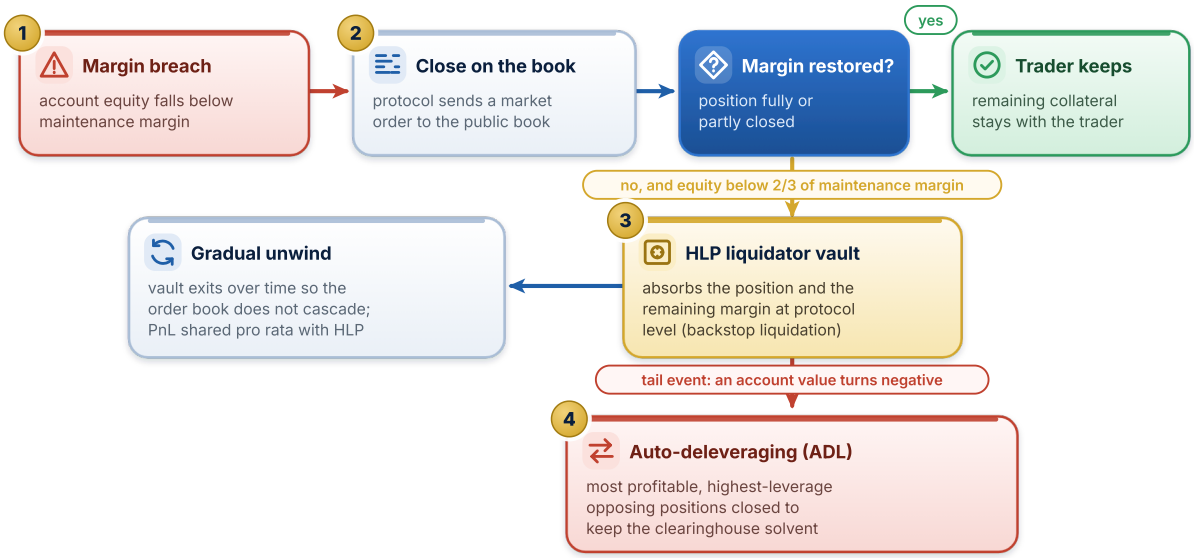


Figure 4. The default waterfall: the public book first, then the HLP liquidator vault, then auto-deleveraging only if an account goes negative.

HLP (Hyperliquid Liquidity Provider)

- **Passive market making:** anyone can deposit USDC. The vault runs automated market-making strategies on-chain, quoting both sides across the order books, and shares profits and losses pro rata with depositors.
- **Backstop liquidations:** if a position cannot be closed on the book, or the account falls past the backstop threshold (2/3 of maintenance margin), the HLP liquidation vault absorbs the position and remaining margin at the protocol level and unwinds it gradually. V10 [1,5]

Auto-deleveraging (ADL)

If a black-swan move wipes out a losing trader's margin and threatens the HLP buffer, a deterministic ADL queue closes the most profitable, highest-leverage opposing positions to keep the clearinghouse solvent. Binance and Bybit use the same type of mechanism.[1,6]

Plain-English version

If you are winning big with high leverage on the other side of a failed trader, you may be closed out automatically, at the mark price, to cover that trader's hole. The protocol prefers this to having bad debt.

WHAT IT MEANS FOR CAPITAL MARKETS

- **Map it to a CCP default waterfall.** Book liquidation resembles a close-out auction; HLP plays a role somewhat like a default fund plus a backstop liquidity provider; ADL resembles loss allocation to surviving members. The key difference: HLP's capital comes from voluntary depositors, not from clearing members with mutualised obligations.
- **HLP depositors take liquidation risk.** Anyone evaluating HLP as an investment is underwriting market making and backstop liquidation, not earning a fixed yield.
- **ADL is a hedging risk.** A hedge on this venue can be closed involuntarily at exactly the moment it is most valuable, which affects how desks size and diversify positions across venues.

SECTION 6 · MONEY OUT

6 How assets exit

Withdrawal reverses the deposit journey, with one design advantage: the trader does not need ETH or HYPE to pay for it. A flat USDC fee covers the validators' Arbitrum gas.[1]

1 USDC flat withdrawal fee

~2–5 min typical end-to-end time

< 1 min exit via HyperEVM and an intent bridge

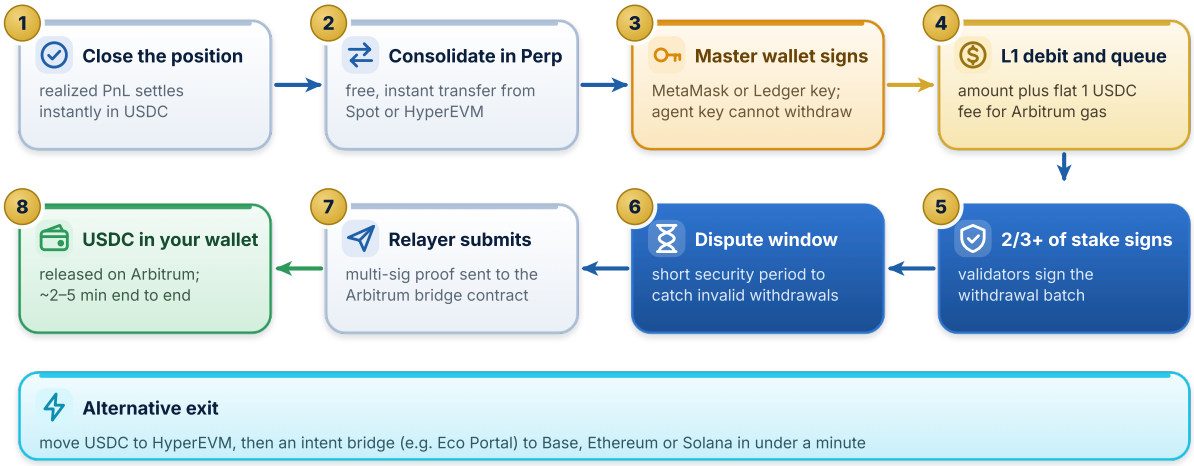


Figure 5. The withdrawal journey via the native Arbitrum bridge, and the faster alternative via HyperEVM and an intent bridge.

1. **Close the position.** Realized PnL settles instantly in USDC into the HyperCore perp balance. USDC on Spot or HyperEVM moves back to Perp with a free, instant transfer.
2. **Sign with the master wallet.** Unlike trading, withdrawal needs the real EVM wallet (for example MetaMask or Ledger). A stolen agent or API key cannot withdraw funds.
3. **L1 deduction.** Hyperliquid deducts the USDC immediately, charges a flat 1 USDC fee to cover Arbitrum gas, and queues the withdrawal.[1,13,14]
4. **Escrow unlock.** Once more than 2/3 of validator stake signs the batch and a short dispute window passes, a relayer submits the multi-signature proof to the Arbitrum bridge contract. Typically 2 to 5 minutes in total. V6
5. **Receipt.** The bridge releases USDC to the trader's Arbitrum address. Alternatively, move USDC to HyperEVM and use an intent bridge such as Eco Portal to reach Base, Ethereum or Solana in under a minute.

WHAT IT MEANS FOR CAPITAL MARKETS

- **Settlement in minutes, not days.** Cash-out takes minutes, not T+1 or T+2, but a dispute window and validator signatures still sit between request and receipt. Operations would treat this as an asynchronous transfer with its own reconciliation step.
- **Key segregation is the main control.** The withdrawal key is the one that matters. For an institution, it belongs in qualified custody with policy controls, while agent keys can be rotated and scoped per desk.
- **Fast exits add counterparties.** Exits through intent bridges are faster but add a solver and another bridge, each with its own settlement and failure modes.

SECTION 7 · VENUE RISK

7 Risks and trade-offs

Hyperliquid's speed comes from deliberate design trade-offs. None of them is unusual for a high-performance chain, but each should be understood before treating it as a venue.[1]

Validator concentration — ELEVATED

WHAT THE BRIEFING SAYS

To process up to 200,000 orders per second with sub-second finality, the validator set is kept smaller and runs on high-spec enterprise hardware, compared with Ethereum's thousands of home nodes.

WHY IT MATTERS

Fewer, larger operators means a lower bar for collusion, outages or coordinated action, and greater reliance on their governance.

Bridge escrow — HIGH

WHAT THE BRIEFING SAYS

The USDC backing the perp exchange sits in a validator-controlled multi-signature contract on Arbitrum. If more than 2/3 of validator keys (by staked HYPE) were compromised or colluded, the escrow would be at risk. V8

WHY IT MATTERS

A single point of failure for collateral. Current docs say most USDC now comes in through Circle's CCTP, which changes the shape of this risk rather than removing it (see V8).

Auto-deleveraging — ELEVATED

WHAT THE BRIEFING SAYS

In extreme tail events, the ADL queue closes the most profitable, highest-leverage opposing positions to keep the clearinghouse solvent.

WHY IT MATTERS

Profitable positions and hedges can be closed involuntarily.

Oracle dependence — ELEVATED

WHAT THE BRIEFING SAYS

Funding and liquidation reference a weighted median of external exchange prices.

WHY IT MATTERS

Venue outages or divergence at source exchanges flow straight into margin outcomes.

Public order flow — MANAGEABLE

WHAT THE BRIEFING SAYS

Orders, fills and positions are fully transparent in real time.

WHY IT MATTERS

Information leakage for large or client-driven flow; positions are visible to other traders.

Reading the colours

HIGH ELEVATED MANAGEABLE

Severity colours are an editorial guide to relative attention in a venue-risk review, not a formal risk rating.

The first three rows come from the briefing text.[1] The last two follow directly from mechanics described in sections 3 and 4.

WHAT IT MEANS FOR CAPITAL MARKETS

- **Venue-risk checklist.** Validator governance, bridge and stablecoin design, oracle method, default waterfall and transparency are the five areas a venue-risk review would cover. Section 8 turns them into questions.
- **Neutral framing.** Each risk is a design choice made to achieve CEX-like speed on a public chain. Whether they are acceptable depends on the use case, such as research, market data, client access or proprietary activity, and on the applicable regulatory perimeter.

SECTION 8 · DUE DILIGENCE

8 Questions to ask

1. **Validators:** who runs them, how is stake distributed, how are new validators admitted, and what happens if a large share goes offline?
2. **Collateral path:** what share of USDC now arrives via native CCTP versus the legacy Arbitrum bridge, and who can pause, upgrade or lock either path?
3. **Oracle:** which exchanges feed the oracle today, with what weights, and what happens when they diverge or one fails?
4. **Default waterfall:** how large is HLP relative to open interest, and how often has ADL been triggered in past stress events?
5. **Keys:** how would we hold the master (withdrawal) key in institutional custody, and how would agent keys be issued, scoped and rotated per desk?
6. **Governance:** how are protocol upgrades proposed, approved and deployed, and with how much notice?
7. **Legal and regulatory:** which legal entity, if any, stands behind the venue, what terms apply, and which regulators consider it in scope for our jurisdictions?
8. **Information leakage:** what would full transparency of orders and positions mean for client confidentiality and our best-execution obligations?

SPOT-CHECKED AGAINST PUBLIC SOURCES

Verification notes

> The briefing text is the backbone of this explainer. The key figures were spot-checked against Hyperliquid's public documentation and other public sources on October 28, 2026. Where the sources differ, the briefing text is kept in the body and the difference is flagged here, not silently changed.

#	Briefing claim	What public sources say	Status
V1	~0.2 s finality	Docs: for a co-located client, median end-to-end order latency is 0.2 s and 99th percentile is 0.9 s. The figure is a median, so the tail is longer.[2]	Consistent (median)
V2	Up to 200,000 orders/sec	Docs: mainnet "currently supports approximately 200k orders/sec"; execution, not consensus, is the bottleneck.[2] This is the protocol's own statement, not an independent benchmark.	Consistent
V3	Blocks every ~100–200 ms	Not stated as a block time on the docs pages checked; the 0.2 s figure is median latency. HyperEVM blocks are slower: small blocks every 1 s and large blocks every 1 minute.[10]	Not confirmed
V4	Minimum deposit 5 USDC	Docs confirm 5 USDC for the Arbitrum bridge. Smaller amounts are not credited and are lost. Credit arrives in under 1 minute.[8]	Confirmed (legacy bridge)
V5	Hourly funding	Confirmed. The rate is computed on an 8-hour basis and one-eighth is paid each hour. The premium is sampled every 5 s, and funding is capped at 4% per hour.[4]	Confirmed
V6	Flat 1 USDC withdrawal fee; 2–5 min	The 1 USDC fee is confirmed by the Hyperliquid community wiki and third-party API docs, but it is not on the official fees page.[13,14] Official docs give 3–4 minutes for the Arbitrum bridge; Chainstack says about 5 minutes.[8,14]	Consistent

V7	Oracle every 3 s from Binance, OKX, Coinbase, Bybit, Kraken	The 3 s update is confirmed. Docs list the sources as Binance, OKX, Bybit, Kraken, KuCoin, Gate.io, MEXC and Hyperliquid spot, weighted 3, 2, 2, 1, 1, 1, 1, 1. **Coinbase is not listed.** Validators' submissions are then combined by a stake-weighted median. The oracle drives funding directly; margining and liquidation use a *mark price* that combines the oracle with Hyperliquid's own book.[3,5]	Differs
V8	All perp USDC sits in the Arbitrum bridge; primary deposit path is that bridge	**Outdated.** Docs say USDC is now natively minted on the Hyperliquid L1 via Circle CCTP, the Arbitrum bridge is "legacy" and deprecated, and it holds less than 10% of USDC on HyperCore.[7,8] Circle notes that HyperCore balances are protocol-level credits, with native USDC held in a Circle contract on HyperEVM.[15] The bridge-escrow risk is therefore smaller in size and partly replaced by CCTP and HyperEVM contract risk.	Differs (material)
V9	HyperEVM contracts trade into HyperCore "within the same block"	Read precompiles reflect the latest HyperCore state when the EVM block is built. But orders and vault transfers sent from HyperEVM via the CoreWriter contract are **deliberately delayed on-chain for a few seconds** so that HyperEVM cannot be used to get a latency advantage.[9]	Differs
V10	HLP backstop below 2/3 of maintenance margin; gradual unwind	The 2/3 threshold is confirmed. Docs add that in a backstop the maintenance margin is not returned to the trader, and positions above 100k USDC are sent to the book 20% at a time. The docs do not describe how the vault unwinds a position.[5]	Partly confirmed
V11	ADL closes most profitable, highest-leverage positions	Confirmed. It triggers when an account's value goes negative. Opposing users are ranked by (mark/entry price) × (notional/account value) and closed at the previous mark price.[6]	Confirmed
V12	CEX ~5–20 ms; hybrid ~10–100 ms latency	Indicative ranges from the briefing; not independently verified.	Not verified

TERMS USED IN THIS EXPLAINER

Glossary

ADL (auto-deleveraging): last-resort mechanism that closes profitable opposing positions to cover an account that has gone negative.

Agent key / API wallet: a session key that can place, modify and cancel orders but cannot withdraw.

AMM (automated market maker): a pool of tokens priced by a formula instead of an order book.

Bridge: a mechanism that locks an asset on one chain and credits a representation on another.

CCTP: Circle's Cross-Chain Transfer Protocol, which burns USDC on one chain and mints it natively on another.

CLOB (central limit order book): a list of buy and sell orders matched by price, then time.

CoreWriter: the HyperEVM system contract that sends actions, such as orders, into HyperCore.

Cross / isolated margin: collateral shared across all positions, or ring-fenced to one position.

Dispute window: a waiting period after a withdrawal is signed, during which invalid withdrawals can be blocked.

Finality: the point at which a transaction cannot be reversed.

Funding rate: a periodic payment between longs and shorts that keeps a perp's price close to the spot price.

HLP: Hyperliquid Liquidity Provider, a community vault that makes markets and backstops liquidations.

HYPE: Hyperliquid's native token, used for staking and HyperEVM gas.

HyperBFT: Hyperliquid's Byzantine-fault-tolerant proof-of-stake consensus, a variant of HotStuff.[2]

HyperCore: the hard-coded trading engine: order books, clearinghouse, liquidations and oracle.

HyperEVM: the general-purpose smart contract environment on the same chain.

Intent bridge / solver: a cross-chain service where a market maker fronts funds on the destination chain and is repaid on the source chain.

Maintenance margin: the minimum equity needed to keep a position open.

Maker / taker: a maker adds a resting order to the book; a taker trades against one.

Mark price: the price used for margining and liquidation, combining the oracle with the exchange's own book.

Oracle price: a weighted median of external exchange prices published by validators.

Perpetual (perp): a futures contract with no expiry, kept near spot by funding payments.

Precompile: a built-in function at a fixed address that contracts can call, here used to read HyperCore state.

Tick / lot size: the minimum price increment and minimum quantity increment for orders.

TWAP: time-weighted average price, an order split into slices over time.

Unit protocol: a guardian network that locks native BTC, ETH or SOL and mints 1:1 tokens on Hyperliquid.

Validator: an operator that stakes HYPE, orders transactions and runs the chain's logic.

RETRIEVED OCTOBER 28, 2026

Sources

Public sources were retrieved on October 28, 2026 (US Eastern time). Protocol documentation changes often; check it again before relying on any parameter.

1. Briefing text on the Hyperliquid model, provided by the Head of Digital Asset Technology, October 28, 2026. This is the content backbone for sections 1–7.
2. Hyperliquid Docs, "HyperCore: Overview" (HyperBFT, a HotStuff variant; median 0.2 s and p99 0.9 s end-to-end latency; ~200k orders/sec). hyperliquid.gitbook.io/hyperliquid-docs/hypercore/overview
3. Hyperliquid Docs, "Oracle" (published every 3 s; weighted median of Binance, OKX, Bybit, Kraken, KuCoin, Gate.io, MEXC, Hyperliquid). hyperliquid.gitbook.io/hyperliquid-docs/hypercore/oracle
4. Hyperliquid Docs, "Funding" (paid hourly; peer-to-peer; 4%/hour cap). hyperliquid.gitbook.io/hyperliquid-docs/trading/funding
5. Hyperliquid Docs, "Liquidations" (book first; liquidator vault below 2/3 of maintenance margin; mark price; partial liquidations). hyperliquid.gitbook.io/hyperliquid-docs/trading/liquidations
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Important notes

This explainer is for internal education. It describes how a third-party protocol says it works and is not an assessment, approval or recommendation of any venue, token or product. Any use by RBC Capital Markets would be subject to the usual risk, compliance, legal and technology review.